Rechecked Derivation of the Tree Plus One-Loop Equations

Local Adiabatic Wick Rotation and Real-Time Nonlocal Memory

The oscillator is

$$S[q]=\int \mathrm{d}t\left[\frac{m}{2}\dot q^2-\frac{m\omega^2}{2}q^2-\frac{\lambda}{24}q^4\right],\qquad S_E[q]=\int \mathrm{d}\tau\left[\frac{m}{2}\dot q^2+\frac{m\omega^2}{2}q^2+\frac{\lambda}{24}q^4\right].\tag{1}$$

Dots denote derivatives with respect to the displayed time variable. The two final results are Eq. (A), the local adiabatic one-loop equation obtained by Wick rotating the Euclidean derivative expansion, and Eq. (B), the causal one-loop equation with the first real-time memory correction obtained from the mode correlator.

1 Euclidean expansion and the one-loop determinant

Split

$$q(\tau)=Q(\tau)+\eta(\tau).\tag{2}$$

The kinetic term expands as

$$\frac{m}{2}(\dot Q+\dot\eta)^2=\frac{m}{2}\dot Q^2+m\dot Q\dot\eta+\frac{m}{2}\dot\eta^2,\qquad \int \mathrm{d}\tau\, m\dot Q\dot\eta=-\int \mathrm{d}\tau\, m\ddot Q\eta,\tag{3}$$

where boundary terms are assumed to vanish. The potential expansion is

$$\begin{aligned}\frac{m\omega^2}{2}(Q+\eta)^2+\frac{\lambda}{24}(Q+\eta)^4&=\frac{m\omega^2}{2}Q^2+\frac{\lambda}{24}Q^4+\left(m\omega^2Q+\frac{\lambda}{6}Q^3\right)\eta\\&\quad+\frac{1}{2}\left(m\omega^2+\frac{\lambda}{2}Q^2\right)\eta^2+\frac{\lambda}{6}Q\eta^3+\frac{\lambda}{24}\eta^4.\end{aligned}\tag{4}$$

Therefore

$$S_E[Q+\eta]=S_E[Q]+\int \mathrm{d}\tau\,\eta E_Q+\frac{1}{2}\int \mathrm{d}\tau\,\eta\mathcal{O}_Q\eta+\int \mathrm{d}\tau\left(\frac{\lambda}{6}Q\eta^3+\frac{\lambda}{24}\eta^4\right),\tag{5}$$

with

$$E_Q=-m\ddot Q+m\omega^2Q+\frac{\lambda}{6}Q^3,\tag{6}$$

$$\mathcal{O}_Q=-m\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}+m\omega^2+\frac{\lambda}{2}Q^2(\tau)=m\left[-\frac{\mathrm{d}^2}{\mathrm{d}\tau^2}+\Omega^2(\tau)\right],\qquad \Omega^2(\tau)=\omega^2+\frac{\lambda}{2m}Q^2(\tau).\tag{7}$$

In the effective-action construction the source cancels the term linear in η. The Gaussian path integral then gives

$$\Gamma_E[Q]=S_E[Q]+\frac{\hbar}{2}\operatorname{Tr}\ln\mathcal{O}_Q+\mathcal{O}(\hbar^2).\tag{8}$$

Since $\operatorname{Tr}\ln\mathcal{O}_Q=\operatorname{Tr}\ln m+\operatorname{Tr}\ln[-\partial_\tau^2+\Omega^2]$, the factor m contributes only a field-independent constant. After also subtracting the $Q=0$ determinant,

$$\Delta\Gamma_E^{(1)}=\frac{\hbar}{2}\operatorname{Tr}\ln[-\partial_\tau^2+\Omega^2]-\frac{\hbar}{2}\operatorname{Tr}\ln[-\partial_\tau^2+\omega^2].\tag{9}$$

2 Derivative expansion of the Euclidean determinant

Let

$$A = -\partial_\tau^2 + \Omega^2(\tau), \qquad V(\tau) = \Omega^2(\tau), \qquad f(\tau,p) = p^2 + V(\tau). \tag{10}$$

The Weyl symbol $R(\tau,p)$ of A^{-1} satisfies

$$f \star R = 1, \tag{11}$$

where, to second adiabatic order,

$$f \star R = fR + \frac{\mathrm{i}}{2}(f_\tau R_p - f_p R_\tau) - \frac{1}{8}\left(f_{\tau\tau}R_{pp} - 2f_{\tau p}R_{p\tau} + f_{pp}R_{\tau\tau}\right) + \mathcal{O}(\partial_\tau^4). \tag{12}$$

With $R = R_0 + R_2 + \cdots$, Eq. (11) gives

$$R_0 = \frac{1}{f}. \tag{13}$$

The first-derivative term vanishes because

$$f_\tau (R_0)_p - f_p (R_0)_\tau = V'\left(-\frac{2p}{f^2}\right) - 2p\left(-\frac{V'}{f^2}\right) = 0. \tag{14}$$

Using $f_{\tau p} = 0$, $f_{pp} = 2$,

$$(R_0)_{pp} = -\frac{2}{f^2} + \frac{8p^2}{f^3}, \qquad (R_0)_{\tau\tau} = -\frac{V''}{f^2} + \frac{2(V')^2}{f^3}, \tag{15}$$

the second-order equation is

$$fR_2 - \frac{1}{8}\left[V''(R_0)_{pp} + 2(R_0)_{\tau\tau}\right] = 0. \tag{16}$$

Thus

$$R_2 = -\frac{V''}{2f^3} + \frac{V''p^2}{f^4} + \frac{(V')^2}{2f^4}. \tag{17}$$

The diagonal Green function is $G_E(\tau,\tau) = \int \frac{\mathrm{d}p}{2\pi} R(\tau,p)$. The needed integrals are

$$\int \frac{\mathrm{d}p}{2\pi}\frac{1}{p^2+\Omega^2} = \frac{1}{2\Omega}, \quad \int \frac{\mathrm{d}p}{2\pi}\frac{1}{f^3} = \frac{3}{16\Omega^5}, \quad \int \frac{\mathrm{d}p}{2\pi}\frac{p^2}{f^4} = \frac{1}{32\Omega^5}, \quad \int \frac{\mathrm{d}p}{2\pi}\frac{1}{f^4} = \frac{5}{32\Omega^7}. \tag{18}$$

Therefore

$$\begin{aligned} G_E(\tau,\tau) &= \frac{1}{2\Omega} - \frac{V''}{16\Omega^5} + \frac{5(V')^2}{64\Omega^7} + \mathcal{O}(\partial_\tau^4) \\ &= \frac{1}{2\Omega} - \frac{\ddot\Omega}{8\Omega^4} + \frac{3\dot\Omega^2}{16\Omega^5} + \mathcal{O}(\partial_\tau^4), \end{aligned} \tag{19}$$

where $V' = 2\Omega\dot\Omega$ and $V'' = 2\dot\Omega^2 + 2\Omega\ddot\Omega$ were used.

The determinant obeys

$$\delta\Delta\Gamma_E^{(1)} = \frac{\hbar}{2}\,\mathrm{Tr}(A^{-1}\delta A) = \frac{\hbar}{2}\int \mathrm{d}\tau\, G_E(\tau,\tau)\delta\Omega^2(\tau). \tag{20}$$

To reconstruct a local Lagrangian, set

$$L_1(\Omega,\dot\Omega) = \frac{1}{2}\Omega + a\frac{\dot\Omega^2}{\Omega^3}. \tag{21}$$

Since $\delta\Omega = \delta(\Omega^2)/(2\Omega)$,

$$\frac{\delta}{\delta\Omega^2}\int \mathrm{d}\tau\, L_1 = \frac{1}{2\Omega}\left(\frac{\partial L_1}{\partial\Omega} - \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial L_1}{\partial\dot\Omega}\right)$$

$$= \frac{1}{4\Omega} - a\frac{\ddot{\Omega}}{\Omega^4} + \frac{3a}{2}\frac{\dot{\Omega}^2}{\Omega^5}. \tag{22}$$

Comparison with $G_E/2$ from Eq. (19) fixes $a = 1/16$. Hence

$$\Delta\Gamma_E^{(1)} = \hbar \int \mathrm{d}\tau \left[\frac{1}{2}(\Omega - \omega) + \frac{\dot{\Omega}^2}{16\Omega^3} + \mathcal{O}(\partial_\tau^4)\right]. \tag{23}$$

Using Eq. (7),

$$2\Omega\dot{\Omega} = \frac{\lambda}{m}Q\dot{Q}, \qquad \dot{\Omega} = \frac{\lambda Q\dot{Q}}{2m\Omega}, \tag{24}$$

and therefore

$$\frac{\dot{\Omega}^2}{16\Omega^3} = \frac{\lambda^2 Q^2\dot{Q}^2}{64m^2\Omega^5}. \tag{25}$$

The local Euclidean effective action is then

$$\Gamma_E[Q] \simeq \int \mathrm{d}\tau \left[\frac{1}{2}Z(Q)\dot{Q}^2 + V_{\mathrm{eff}}(Q)\right], \tag{26}$$

with

$$V_{\mathrm{eff}}(Q) = \frac{m\omega^2}{2}Q^2 + \frac{\lambda}{24}Q^4 + \frac{\hbar}{2}[\Omega(Q) - \omega], \qquad \Omega(Q) = \sqrt{\omega^2 + \frac{\lambda}{2m}Q^2}, \tag{27}$$

$$Z(Q) = m + \hbar\frac{\lambda^2 Q^2}{32m^2\Omega^5}. \tag{28}$$

3 Local equation and Wick rotation

For

$$L_E = \frac{1}{2}Z(Q)\,\dot{Q}^2 + V_{\mathrm{eff}}(Q), \tag{29}$$

one has

$$\frac{\partial L_E}{\partial Q} = \frac{1}{2}Z'(Q)\dot{Q}^2 + V'_{\mathrm{eff}}(Q), \qquad \frac{\partial L_E}{\partial \dot{Q}} = Z(Q)\,\dot{Q}, \tag{30}$$

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial L_E}{\partial \dot{Q}} = Z(Q)\ddot{Q} + Z'(Q)\dot{Q}^2. \tag{31}$$

Thus $\partial L_E/\partial Q - \mathrm{d}(\partial L_E/\partial\dot{Q})/\mathrm{d}\tau = 0$ gives

$$-Z(Q)\ddot{Q} - \frac{1}{2}Z'(Q)\dot{Q}^2 + V'_{\mathrm{eff}}(Q) = 0. \tag{32}$$

The derivative of Ω with respect to Q is

$$\Omega'(Q) = \frac{\lambda Q}{2m\Omega}. \tag{33}$$

Therefore

$$V'_{\mathrm{eff}}(Q) = m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\hbar}{2}\Omega'(Q) = m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\Omega}. \tag{34}$$

Also,

$$\begin{aligned} Z'(Q) &= \frac{\hbar\lambda^2}{32m^2}\frac{\mathrm{d}}{\mathrm{d}Q}\left(Q^2\Omega^{-5}\right) \\ &= \frac{\hbar\lambda^2}{32m^2}\left[2Q\Omega^{-5} - 5Q^2\Omega^{-6}\Omega'(Q)\right] \\ &= \frac{\hbar\lambda^2}{32m^2}\left[\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right]. \end{aligned} \tag{35}$$

Now rotate the already local equation by

$$\tau = \mathrm{i}t, \qquad \frac{\mathrm{d}Q}{\mathrm{d}\tau} = -\mathrm{i}\frac{\mathrm{d}Q}{\mathrm{d}t}, \qquad \frac{\mathrm{d}^2Q}{\mathrm{d}\tau^2} = -\frac{\mathrm{d}^2Q}{\mathrm{d}t^2}. \tag{36}$$

Equation (32) becomes

$$Z(Q)\ddot{Q} + \frac{1}{2}Z'(Q)\dot{Q}^2 + V'_{\text{eff}}(Q) = 0, \tag{37}$$

where dots now denote $\mathrm{d}/\mathrm{d}t$. Substituting Eqs. (28), (34), and (35), the local adiabatic tree plus one-loop equation is

$$\boxed{\begin{aligned} 0 = {} & \left[m + \hbar\frac{\lambda^2 Q^2}{32m^2\Omega^5}\right]\ddot{Q} + \frac{\hbar\lambda^2}{64m^2}\left[\frac{2Q}{\Omega^5} - \frac{5\lambda Q^3}{2m\Omega^7}\right]\dot{Q}^2 \\ & + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\Omega}, \qquad \Omega = \sqrt{\omega^2 + \frac{\lambda}{2m}Q^2}. \end{aligned}} \tag{A}$$

The derivative expansion assumes

$$\frac{|\dot{\Omega}|}{\Omega^2} \ll 1, \qquad \frac{|\ddot{\Omega}|}{\Omega^3} \ll 1, \tag{38}$$

and omits $\mathcal{O}(\hbar\partial_t^4)$ and $\mathcal{O}(\hbar^2)$ terms. Equation (A) is conservative because it follows from

$$\Gamma^{\text{loc}}_{\text{RT}} = \int \mathrm{d}t \left[\frac{1}{2}Z(Q)\dot{Q}^2 - V_{\text{eff}}(Q)\right]. \tag{39}$$

4 Real-time correlator from mode averaging

The Heisenberg equation is

$$m\ddot{q} + m\omega^2 q + \frac{\lambda}{6}q^3 = 0. \tag{40}$$

Write $q = Q + \eta$ with $Q = \langle q\rangle$, $\langle\eta\rangle = 0$. Since

$$\langle (Q+\eta)^3\rangle = Q^3 + 3Q\langle\eta^2\rangle + \langle\eta^3\rangle, \tag{41}$$

and $\langle\eta^3\rangle = 0$ in the Gaussian one-loop approximation, the one-loop tadpole equation is

$$0 = m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \frac{\lambda}{2}Q(t)\langle\eta^2(t)\rangle. \tag{42}$$

The quadratic real-time action for η is

$$S_2[\eta; Q] = \frac{m}{2}\int \mathrm{d}t \left[\dot{\eta}^2 - \Omega_Q^2(t)\eta^2\right], \qquad \Omega_Q^2(t) = \omega^2 + \delta\Omega^2(t), \qquad \delta\Omega^2(t) = \frac{\lambda}{2m}Q^2(t). \tag{43}$$

Thus each fluctuation mode obeys the oscillator equation with time-dependent frequency. For the single quantum-mechanical oscillator there is one mode; in field theory the replacement is a sum/integral over momenta. Use

$$\hat{\eta}(t) = \sqrt{\frac{\hbar}{2m}}\left[a u(t) + a^\dagger u^*(t)\right], \qquad [a, a^\dagger] = 1. \tag{44}$$

The equation of motion and Wronskian normalization are

$$\ddot{u} + \Omega_Q^2(t)u = 0, \qquad W[u] \equiv u\dot{u}^* - \dot{u}u^* = 2\mathrm{i}. \tag{45}$$

The Wronskian gives the canonical commutator because

$$[\hat{\eta}, m\dot{\hat{\eta}}] = \frac{\hbar}{2}\left(u\dot{u}^* - u^*\dot{u}\right) = \mathrm{i}\hbar. \tag{46}$$

In the state annihilated by a,

$$\langle \eta^2(t) \rangle = \frac{\hbar}{2m} |u(t)|^2. \tag{47}$$

Equations (42), (45), and (47) give the exact causal one-loop backreaction system. The next steps compute its first explicit nonlocal correction.

Choose the free positive-frequency initial mode

$$u_0(t) = \frac{e^{-\mathrm{i}\omega(t-t_0)}}{\sqrt{\omega}}, \qquad \ddot{u}_0 + \omega^2 u_0 = 0, \qquad u_0 \dot{u}_0^* - \dot{u}_0 u_0^* = 2\mathrm{i}. \tag{48}$$

Write $u = u_0 + \delta u$. To first order in $\delta\Omega^2$,

$$(\partial_t^2 + \omega^2)\delta u(t) = -\delta\Omega^2(t) u_0(t). \tag{49}$$

The retarded Green function of $\partial_t^2 + \omega^2$ is

$$G_R(t-t') = \theta(t-t') \frac{\sin[\omega(t-t')]}{\omega}, \qquad (\partial_t^2 + \omega^2) G_R(t-t') = \delta(t-t'). \tag{50}$$

Therefore

$$\delta u(t) = -\int_{t_0}^{t} \mathrm{d}t' \, \frac{\sin[\omega(t-t')]}{\omega} \, \delta\Omega^2(t') u_0(t'). \tag{51}$$

Let $\Delta = t - t'$. Then

$$u_0^*(t)\delta u(t) = -\int_{t_0}^{t} \mathrm{d}t' \, \frac{\sin(\omega\Delta)}{\omega^2} e^{\mathrm{i}\omega\Delta} \delta\Omega^2(t'), \tag{52}$$

and adding the complex conjugate gives

$$\begin{aligned} |u(t)|^2 &= |u_0|^2 + u_0^* \delta u + u_0 \delta u^* + \mathcal{O}((\delta\Omega^2)^2) \\ &= \frac{1}{\omega} - \frac{2}{\omega^2} \int_{t_0}^{t} \mathrm{d}t' \, \sin(\omega\Delta)\cos(\omega\Delta)\delta\Omega^2(t') + \mathcal{O}((\delta\Omega^2)^2) \\ &= \frac{1}{\omega} - \frac{1}{\omega^2} \int_{t_0}^{t} \mathrm{d}t' \, \sin[2\omega(t-t')]\delta\Omega^2(t') + \mathcal{O}((\delta\Omega^2)^2). \end{aligned} \tag{53}$$

Using $\delta\Omega^2 = \lambda Q^2/(2m)$,

$$\langle \eta^2(t) \rangle = \frac{\hbar}{2m\omega} - \frac{\hbar\lambda}{4m^2\omega^2} \int_{t_0}^{t} \mathrm{d}t' \, \sin[2\omega(t-t')] Q^2(t') + \mathcal{O}\left(\frac{\hbar}{m} \frac{(\delta\Omega^2)^2}{\omega^5} \right). \tag{54}$$

Substitution into Eq. (42) gives the explicit causal nonlocal tree plus one-loop equation

$$\boxed{\begin{aligned} 0 = {}& m\ddot{Q}(t) + m\omega^2 Q(t) + \frac{\lambda}{6} Q^3(t) + \hbar \frac{\lambda}{4m\omega} Q(t) \\ & - \hbar \frac{\lambda^2}{8m^2\omega^2} Q(t) \int_{t_0}^{t} \mathrm{d}t' \, \sin[2\omega(t-t')] Q^2(t') + \mathcal{O}\left(\frac{\hbar\lambda Q}{m} \frac{(\delta\Omega^2)^2}{\omega^5} \right). \end{aligned}} \tag{B}$$

For finite t_0, Eq. (B) contains physical initial-time transients. To compare with the adiabatic effective action one takes the usual retarded adiabatic-switching limit, equivalently $t_0 \to -\infty$ with a factor $e^{-\epsilon(t-t')}$, $\epsilon \to 0^+$, in the memory integral.

5 Local limit and coefficient check

Define the adiabatically switched memory functional

$$I[f](t) = \lim_{\epsilon\to 0^+} \int_0^\infty \mathrm{d}s\, e^{-\epsilon s} \sin(2\omega s) f(t-s), \qquad f(t) = Q^2(t). \tag{55}$$

For slowly varying f,

$$f(t-s) = f(t) - s\dot{f}(t) + \frac{s^2}{2}\ddot{f}(t) + \cdots . \tag{56}$$

The regulated moments are

$$\int_0^\infty \mathrm{d}s\, e^{-\epsilon s} \sin(as) = \frac{a}{a^2+\epsilon^2} \xrightarrow{\epsilon\to 0^+} \frac{1}{a}, \tag{57}$$

$$\int_0^\infty \mathrm{d}s\, s e^{-\epsilon s} \sin(as) = \frac{2a\epsilon}{(a^2+\epsilon^2)^2} \xrightarrow{\epsilon\to 0^+} 0, \tag{58}$$

$$\int_0^\infty \mathrm{d}s\, s^2 e^{-\epsilon s} \sin(as) = \frac{2a(3\epsilon^2 - a^2)}{(a^2+\epsilon^2)^3} \xrightarrow{\epsilon\to 0^+} -\frac{2}{a^3}. \tag{59}$$

With $a = 2\omega$,

$$I[Q^2](t) = \frac{Q^2}{2\omega} - \frac{1}{(2\omega)^3}\frac{\mathrm{d}^2 Q^2}{\mathrm{d}t^2} + \mathcal{O}(\partial_t^4). \tag{60}$$

The memory force in Eq. (B) therefore becomes

$$\begin{aligned} -\hbar\frac{\lambda^2}{8m^2\omega^2} Q I[Q^2] &= -\hbar\frac{\lambda^2 Q^3}{16m^2\omega^3} + \hbar\frac{\lambda^2}{64m^2\omega^5} Q\frac{\mathrm{d}^2Q^2}{\mathrm{d}t^2} + \mathcal{O}(\partial_t^4) \\ &= -\hbar\frac{\lambda^2 Q^3}{16m^2\omega^3} + \hbar\frac{\lambda^2}{32m^2\omega^5}\left(Q^2\ddot{Q} + Q\dot{Q}^2\right) + \mathcal{O}(\partial_t^4), \end{aligned} \tag{61}$$

because

$$Q\frac{\mathrm{d}^2Q^2}{\mathrm{d}t^2} = Q\frac{\mathrm{d}}{\mathrm{d}t}(2Q\dot{Q}) = 2Q^2\ddot{Q} + 2Q\dot{Q}^2. \tag{62}$$

Now expand the adiabatic equation (A) to the same order. Since

$$\frac{1}{\Omega} = \frac{1}{\omega}\left(1 - \frac{\lambda Q^2}{4m\omega^2} + \mathcal{O}(\lambda^2 Q^4)\right), \qquad \frac{1}{\Omega^5} = \frac{1}{\omega^5} + \mathcal{O}(\lambda Q^2), \tag{63}$$

Eq. (A) gives, through $\mathcal{O}(\hbar\lambda^2)$,

$$0 = m\ddot{Q} + m\omega^2 Q + \frac{\lambda}{6}Q^3 + \hbar\frac{\lambda Q}{4m\omega} - \hbar\frac{\lambda^2 Q^3}{16m^2\omega^3} + \hbar\frac{\lambda^2}{32m^2\omega^5}\left(Q^2\ddot{Q} + Q\dot{Q}^2\right) + \cdots . \tag{64}$$

This is exactly the local derivative expansion of Eq. (B) in the adiabatic-switching limit. The recheck also shows why the $\dot{Q}^2$ term in Eq. (A) is not friction: it is part of the conservative local expansion of the causal memory kernel.